

COLLINS BAFFOUR KYEI

Economist | Doctoral Researcher | University Lecturer | Applied Econometrician

School of Economics, University of Cape Coast, Ghana

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ACADEMIC PROFILE

Economist, doctoral researcher and university lecturer specialising in applied econometrics, monetary and financial economics, and machine-learning applications to economic research. Published in international peer-reviewed journals and experienced in postgraduate and undergraduate teaching across data analysis, quantitative research methods, econometrics and policy modelling. Contributing writer to a national values education handbook for Senior High School students in Ghana and to university-level economics teaching texts. Brings together advanced quantitative research, research software development, academic programme coordination, data-literacy capacity building and multi-stakeholder public engagement.

ACADEMIC HIGHLIGHTS

4 peer-reviewed journal articles	MSc lead lecturer for 3-credit Data Analysis	4 educational text and handbook contributions
400+ online learners supported through programme team	130+ SHS students reached through mentorship programme	500+ conference participants supported through organising team

EDUCATION

Doctor of Philosophy (PhD) in Economics - In Progress

University of Cape Coast, Ghana | 2023-Present

- Research focus: machine-learning and advanced econometric applications to inflation-related issues in Ghana.
- Doctoral training in macroeconomics, econometric theory, policy analysis and advanced empirical research.

Master of Philosophy (MPhil) in Economics

University of Cape Coast, Ghana | 2021-2023

- Thesis focus: interconnectedness among commodity prices, banking-sector financial soundness indicators and selected macroeconomic variables in Ghana.
- Methods included quantile regression, non-parametric causality, bi-wavelet and partial wavelet analysis, TVP-VAR connectedness and wavelet-multiple techniques.

Bachelor of Commerce (Management)

University of Cape Coast, Ghana | 2015-2019

- Project: Effect of Career Development on Employee Engagement among Lecturers in the College of Humanities and Legal Studies, University of Cape Coast.

Certificate Programme for Doctoral Studies

BAUM Tenpers Institute | 2023-2024

- Preliminary doctoral coursework in Mathematics, Econometric Theory, Microeconomic Theory and Macroeconomic Theory.

CURRENT ACADEMIC APPOINTMENTS

Graduate Assistant

Department of Data Science and Economic Policy, School of Economics, University of Cape Coast | 2023-Present

- Teach and support economics, econometrics, computing and postgraduate policy/data-analysis courses; provide student research and software guidance; and support CDAMAA data activities.

Lecturer - DIS 814: Data Analysis

Department of Information Science, Faculty of Arts, University of Cape Coast | 2025/2026, Second Semester

- Lead a 3-credit MSc course with 27 scheduled contact hours covering qualitative and quantitative analysis, NVivo, IBM SPSS Statistics, SmartPLS and data ethics.

ACADEMIC APPOINTMENTS AND TEACHING EXPERIENCE

Graduate Assistant

Department of Data Science and Economic Policy, School of Economics, University of Cape Coast | 2023-Present

- Deliver undergraduate instruction in Project Appraisal I and II, Introduction and Intermediate Computing for Social Scientists, and Economics of Capital Markets.
- Contribute to instruction and applied sessions in Intermediate Statistics, Principles of Microeconomics and Macroeconomics, Cross-Sectional and Panel Econometrics, Time Series Econometrics and Financial Econometrics.
- Support postgraduate teaching in Economic Policy Modelling, Public Policy Modelling, Data Literacy and Mining, and Time Series Econometrics.
- Develop lecture materials, practical datasets, analytical exercises and assessment resources; provide methodological and statistical-software guidance to students.
- Support data archiving, cleaning, documentation and analytical work through the Centre for Data Archiving, Management, Analysis and Advocacy (CDAMAA).

Lecturer - DIS 814: Data Analysis (3 Credits)

Department of Information Science, Faculty of Arts, University of Cape Coast | 2025/2026, Second Semester

- Serve as lead lecturer for a postgraduate Data Analysis course for MSc Information Science students, combining conceptual teaching with hands-on computer laboratory sessions.
- Designed and deliver the course across the full data-analysis pipeline: data collection, management and cleaning; qualitative coding and modelling; quantitative analysis; structural equation modelling; and ethical data practice.
- Teach using IBM SPSS Statistics, NVivo and SmartPLS, with emphasis on methodological justification, software execution and interpretation rather than button-clicking.
- Cover qualitative coding and content analysis, descriptive and inferential statistics, reliability analysis, t-tests, ANOVA, correlation, chi-square and PLS-SEM.
- Design practical datasets, assignments and analytical exercises, and mentor students in selecting methods appropriate to dissertation-level research questions.

Graduate Assistant - Quantitative Research Methods

Department of Sociology and Anthropology, Faculty of Social Sciences, University of Cape Coast | 2025/2026, Second Semester

- Co-deliver postgraduate quantitative research methods instruction for social science students transitioning from predominantly qualitative backgrounds to applied quantitative research.
- Facilitate SPSS and SmartPLS practical sessions covering data preparation, assumption checking, hypothesis testing and empirical interpretation.
- Support instruction on t-tests, ANOVA, ANCOVA, MANCOVA, Pearson and Spearman correlation, chi-square, OLS multiple regression and binary logistic regression.
- Guide students through measurement theory, reliability and validity, Exploratory Factor Analysis, sampling, questionnaire development and the conceptual foundations of PLS-SEM.
- Provide methodological guidance for quantitative research designs, proposals and APA 7 reporting of empirical results.

Part-Time Lecturer

Department of Accountancy and Finance, Cape Coast Technical University | 2023-May 2025

- Delivered undergraduate instruction in economics and finance, integrating applied quantitative examples and contemporary empirical evidence into teaching.
- Designed course materials, assessments and analytical exercises aligned with programme learning outcomes.
- Provided academic advising, feedback and mentorship in research, academic progression and career development.
- Contributed to departmental activities and the integration of digital teaching methods into course delivery.

Teaching Assistant - National Service Personnel

Department of Management, School of Business, University of Cape Coast | 2019-2020

- Facilitated tutorials in International Business, Innovation and Technology Management, Strategic Management, Principles of Management and Managerial Economics.
- Worked with faculty to prepare lecture notes and convert course content into student-friendly presentation materials.
- Supported coursework, presentations, quizzes and examinations, and provided student research and career mentoring.

SELECTED TEACHING PORTFOLIO

Course / Subject Area	Level	Primary Teaching Contribution
DIS 814: Data Analysis	MSc	Lead course delivery; NVivo, SPSS, SmartPLS, qualitative and quantitative analysis
Quantitative Research Methods	Postgraduate	SPSS, EFA, MANCOVA, regression, logistic regression, PLS-SEM and research design
Economic Policy Modelling	Postgraduate	Teaching support and applied policy modelling
Public Policy Modelling	Postgraduate	Teaching support and quantitative policy analysis
Data Literacy and Mining	Postgraduate	Teaching support and data-literacy applications
Time Series Econometrics	Postgraduate / Undergraduate	Teaching support and empirical time-series methods
Project Appraisal I & II	Undergraduate	Lecture delivery and applied project evaluation
Computing for Social Scientists	Undergraduate	Lecture delivery and practical computing instruction
Economics of Capital Markets	Undergraduate	Lecture delivery
Cross-Sectional & Panel Econometrics	Undergraduate	Teaching and practical support
Financial Econometrics	Undergraduate	Teaching and practical support

RESEARCH PROFILE AND METHODOLOGICAL EXPERTISE

RESEARCH THEMES

Inflation dynamics; monetary policy; central banking; financial stability; financial economics; financial technologies; economic policy uncertainty; public finance; corporate governance; managerial economics.

METHODS AND ANALYTICS

Quantile regression and quantile-on-quantile methods; non-parametric causality; bi-wavelet and partial wavelet coherence; wavelet-multiple techniques; TVP-VAR connectedness; time-series and panel econometrics; machine learning and ensemble methods; SHAP explainability; bibliometric analysis; systematic literature review.

RESEARCH SOFTWARE AND OPEN METHODS

qqr - R package for Quantile-on-Quantile Regression

Package developer and maintainer | 2026

- Developing and preparing an R package that supports quantile-on-quantile regression workflows, analytical output and visualisation for applied researchers.
- Presented the package and its methodological motivation in a 2026 research seminar titled “Measuring What Standard Models Miss: Quantile-on-Quantile Regression and the qqr Package for R.”

PEER-REVIEWED PUBLICATIONS

1. Kyei, C. B., Armah, M. K., Cantah, W. G., & Appiah, E. S. (2026). Inflation trajectory in Ghana from dawn to dusk: A bibliometric analysis and systematic literature review. *Social Sciences & Humanities Open*, 13, 102538. <https://doi.org/10.1016/j.ssaho.2026.102538>
2. Kyei, C. B., Ampong, G. O. A., Owusu Junior, P., Simpe Ofori, K., David N'Dri, K., & N'Da, K. (2024). Information flow between BRVM and ESG stock returns: A frequency-dependent analysis. *Research in Globalization*, 8, 100192. <https://doi.org/10.1016/j.resglo.2024.100192>
3. Kyei, C. B., Cantah, W. G., & Owusu Junior, P. (2023). Effect of commodity prices on financial soundness: Insight from adaptive market hypothesis in the Ghanaian setting. *Resources Policy*, 86, 104076. <https://doi.org/10.1016/j.resourpol.2023.104076>
4. Kyei, C. B., Asafo-Adjei, E., Owusu Junior, P., Adam, A. M., Idun, A. A.-A. A., & Boateng, J. K. G. A. G. A. (2023). Nexus between commodities and banking sector financial soundness: The role of general macroeconomic setting in Ghana. *Cogent Economics & Finance*, 11(1), 2173871. <https://doi.org/10.1080/23322039.2023.2173871>

MANUSCRIPTS UNDER REVIEW

1. Kyei, C. B., Armah, M. K., Cantah, W. G., & Appiah, E. S. (2026). From Barrel to Basket: Which Oil Shock Drives Inflation in West Africa?
2. Kyei, C. B., Armah, M. K., Cantah, W. G., & Appiah, E. S. (2026). Conditional Fisher Effects in a Frontier Inflation-Targeting Economy: Time-Frequency and Quantile Evidence from Ghana.
3. Kyei, C. B., Armah, M. K., Cantah, W. G., & Appiah, E. S. (2026). Exchange Rate, Global Uncertainty, and Producer-Consumer Price Co-movement in Ghana: A Wavelet Coherence Analysis.
4. Dasmani, I., Alhassan, A.-W. K., Abokyi, E., & Kyei, C. B. (2026). Livelihood Shocks and Household Recovery in Rural Ghana: Evidence from Apptatse.
5. Dasmani, I., Cantah, W. G., Boakye, D. C., Osei, C., & Kyei, C. B. (2026). Predicting High Inflation Episodes Across West Africa: Using Machine Learning, Ensemble Methods, and SHAP Explainability.

TEXTBOOKS AND EDUCATIONAL PUBLICATIONS

Selected writing contributions. Full bibliographic details should be added when final title-page, edition, publisher and ISBN information is available.

Essential Values for Ghanaian Youth/Students Handbook

Contributing Writer | National SHS values education resource

- Contributed to educational content designed for Senior High School students across Ghana, promoting integrity, respect, responsibility, patriotism, critical thinking and empathy as foundations for citizenship and social cohesion.

Principles of Microeconomics

Contributing Writer | University of Cape Coast teaching text

- Contributed to student-centred instructional content used in introductory economics teaching for early-year University of Cape Coast students.

Principles of Macroeconomics

Contributing Writer | University of Cape Coast teaching text

- Contributed to the development of foundational macroeconomics content for undergraduate economics instruction.

Elements of Macroeconomics

Contributing Writer | University of Cape Coast teaching text

- Supported the preparation of accessible macroeconomic theory, concepts and applications for first- and second-year economics students.

ACADEMIC PROGRAMME COORDINATION, CAPACITY BUILDING AND OUTREACH

Programme Organising and Implementation Team Member - UCC Data Literacy Week 2026

Department of Data Science and Economic Policy, University of Cape Coast | Bank of Ghana sponsored | 2026

- Contributed to planning and implementing a six-component data-literacy and public-engagement programme under the theme "Building a Data-Smart Ghana: Evidence, Ethics and Inclusion."
- Supported stakeholder consultation, teacher and mentor capacity building, online Data Science and AI training, SHS mentorship, community outreach and a national stakeholders conference.
- Contributed to an initiative that trained 400+ online participants, reached 130+ SHS students across seven schools, mobilised 12 UCC mentors, engaged artisans and traders, and convened 500+ stakeholders at the final conference.
- Supported practical data-literacy activities promoting evidence-based decision-making, responsible data use and inclusive digital capacity building.

Programme Coordination Team Member - Artificial Intelligence in Higher Education Workshop

University of Cape Coast / Professors Without Borders | 15-19 June 2026

- Served on the UCC Coordination Team for a five-day international capacity-building programme on the responsible integration of Artificial Intelligence into higher education.
- Supported facilitator reception, campus and institutional orientation, workshop logistics and engagement with lecturers, graduate assistants and students.
- Contributed to the coordination of sessions covering andragogy, experiential learning, TPACK, SAMR, AI ethics, cognitive challenges, Design Thinking and AI-supported entrepreneurship.
- Supported academic knowledge exchange and institutional engagement linked to Professors Without Borders.

Organising and Programme Support Team Member - Bayesian Networks and Impact Evaluation in Economics

University of Cape Coast / Professors Without Borders | 22-26 June 2026

- Supported the organisation and delivery of a five-day intensive quantitative-methods workshop for faculty, researchers and graduate students.
- Contributed to programme coordination for training in multinomial and Gaussian Bayesian Networks, Dynamic Bayesian Networks, the Kalman Filter and modern causal impact evaluation.
- Supported an R-based programme covering RCTs, Propensity Score Matching, Difference-in-Differences and Regression Discontinuity Design for approximately 46-48 participants.
- Contributed to institutional capacity building in causal inference, probabilistic reasoning and applied economic research.

Organising Support - Equipping Women for Impact in Teaching, Research and Leadership

University of Cape Coast / NORHED-funded ECOGIV initiative | December 2025

- Supported the organisation and implementation of a capacity-building programme for female academics.
- Contributed to workshop support around research data management, open science, digital teaching tools, publication planning and inclusive academic leadership.
- Supported CDAMAA and departmental facilitators in programme coordination and workshop delivery.

Academic Workshop and Seminar Facilitation

University of Cape Coast | 2024-2025

- Supported the organisation and facilitation of the GRASAG-UCC two-day Research Training Workshop in December 2024.
- Contributed to the Data Science, Analytics and Academic Research seminar/training series conducted between May and July 2025.

RESEARCH AND PROFESSIONAL EXPERIENCE

Volunteer Research Assistant to Dr William Godfred Cantah

Department of Data Science and Economic Policy, University of Cape Coast | 2023-Present

- Conduct literature synthesis and contribute to research design, methodological specification and empirical strategy development in applied economics and policy research.
- Support data collection, cleaning, documentation and statistical analysis using R, Stata, EViews, SPSS and other analytical tools.

- Contribute to model estimation, results interpretation, manuscript development, report writing and preparation of research outputs for peer review.
- Maintain research datasets, analytical documentation and reference materials to support reproducibility and collaborative research workflows.

Research Committee Chair

National Union of Ghana Students | 2023-2024

- Led student research initiatives aligned with the Union's development and policy objectives.
- Reviewed research proposals, methodologies and outputs; supported analytical quality and relevance.
- Managed research-team activities, delegated assignments and mentored committee members.
- Compiled reports and presented analytical findings to NUGS leadership and relevant stakeholders.

Volunteer Research Assistant to Dr Peterson Owusu Junior

Department of Finance, School of Business, University of Cape Coast | 2020-2023

- Conducted literature reviews and supported empirical finance research across commodity, macroeconomic and market-related topics.
- Collected, organised and analysed data using EViews, R, MATLAB, Stata and SmartPLS.
- Supported conference and research presentations, analytical interpretation and manuscript preparation.
- Applied ethical research practices and continuously developed expertise in emerging quantitative and financial research methods.

SELECTED NATIONAL DATA, RESEARCH AND DEVELOPMENT PROJECTS

Integrated Business Establishment Survey II (IBES II) - Regional Facilitator

Ghana Statistical Service | 15 June-6 July 2025

- Coordinated and oversaw regional survey implementation, supporting alignment with national operational objectives.
- Supervised training and fieldwork processes, monitored data-quality requirements and addressed operational issues.
- Served as liaison for progress reporting, publicity support and return/accountability of survey materials.
- Promoted teamwork, confidentiality and ethical standards among field personnel.

Essential Values for Ghanaian Youth - Vice Board Chair, Youth Advisory Board and Contributing Writer

Ghana Education Service / NUGS / Lead for Ghana / Honour Ghana / T-TEL Ghana | 2023-Present

- Serve as Vice Board Chair of the Youth Advisory Board supporting implementation of the national values-education initiative.
- Contributed to writing and developing the Essential Values handbook intended for Senior High School students across Ghana.
- Supported educational content and youth engagement focused on integrity, respect, responsibility, patriotism, critical thinking and empathy.
- Contributed to student and emerging-writer recruitment, programme implementation and stakeholder engagement.

Ghana Census of Agriculture - Data Validator and Analyst

Ghana Statistical Service | November 2022-May 2023

- Created and formatted analytical tables for the Ghana Census of Agriculture report and validated regional and district outputs.
- Prepared summaries of trends and analytical insights using Stata and Microsoft Excel to support evidence-based interpretation and reporting.

2021 Population and Housing Census Data - Data Validation

Ghana Statistical Service | 13-14 August 2022

- Analysed and replicated tables from the Population and Housing Census report using Stata as part of a focused data-validation exercise.

ACADEMIC SERVICE - JOURNAL PEER REVIEW

Voluntary reviewer of manuscripts submitted to international peer-reviewed journals:

Journal	Year(s)
International Journal of Disaster Resilience in the Built Environment	2026
North American Journal of Economics and Finance	2026
Resources Policy	2023, 2024
Humanities & Social Sciences Communications	2023

CONFERENCE PRESENTATIONS AND ACADEMIC ENGAGEMENT

Measuring What Standard Models Miss: Quantile-on-Quantile Regression and the qqr Package for R

Research Seminar | Presenter | 2026

Second Annual International Virtual Data Science and Analytics Conference

Participant | 2025

5th International Conference of Postgraduate Studies, University of Cape Coast

Rapporteur | 2025

Drivers of E-learning Success and Performance amongst Distance Education Students: The Role of E-learning Satisfaction

Virtual Mobility for All Conference | Poster Presentation | 2023

LEADERSHIP AND INSTITUTIONAL ENGAGEMENT

Role	Organisation	Period
Vice Board Chair	National Youth Advisory Board	2024-Present
Director of Research	Graduate Students Association of Ghana, UCC	2024-2025
Research Committee Chairperson	National Union of Ghana Students	2023-2024
Representative, Global Executive Council	AKATAKYIE BQs Group	2021-Present
Deputy Sponsorship Coordinator	National Service Personnel Association, Cape Coast	2019-2020
Alumni Representative, 2015 Year Group	UCC Department of Management Studies	2019-Present
Financial Secretary	UCC Association of Business Students	2018-2019
Planning Committee Member	UCC School of Business Dinner Planning Committee	2018-2019
Programme Representative, BCom Management 2015 Year Group	University of Cape Coast	2015-2019

TECHNICAL, DATA ANALYSIS AND RESEARCH SKILLS

Area	Advanced / Strong Proficiency	Working / Developing Proficiency
Econometrics and statistics	R; EViews; Stata; IBM SPSS Statistics; SmartPLS; VOSviewer	MATLAB; Python
Research and analytics	Empirical research; quantitative analysis; econometric modelling; data cleaning; literature synthesis; report writing; bibliometric analysis; systematic literature review	Qualitative research; mixed-method integration; NVivo instructional use
Reproducible and computational work	R package development; reproducible analytical workflows; data documentation; research-data management	Cloud-computing concepts; database fundamentals
Digital and productivity tools	Microsoft Office Suite; Mendeley; Google Forms; Generative AI prompt engineering	Canva; Camtasia
Teaching and academic practice	Postgraduate data analysis; quantitative methods; software-based practical instruction; assessment design/support; student mentoring; educational content development	Digital pedagogy and AI-integrated teaching design

PROFESSIONAL TRAINING AND CERTIFICATIONS

Data Science and Analytics Summer Bootcamp

University of Calgary | 2025

African Economic Research Consortium (AERC) Collaborative PhD Programme in Economics

2024

- Public Sector Economics I & II; Econometrics Theory and Practice I & II; Comprehensive Microeconomics Examination; Comprehensive Macroeconomics Examination.

Cloud Computing Training

2023

African Economic Research Consortium (AERC) Collaborative Masters Programme in Economics

2021

- Financial Economics I & II; Econometrics Theory and Practice I & II.

Introduction to Modern Database Systems

Saylor Academy | 2018

Certificate in Microsoft Office 2010

ALISON | 2018

REFEREES

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